

Final Report – Discrete-Event Simulation of Polling Systems

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Repository: garmaar/project3-polling-systems

Primary Analysis Notebook: notebooks/report_analysis.ipynb

1. Executive Summary

This report presents a discrete-event simulation of polling systems under exhaustive and limited service disciplines, implemented in Python. The simulation reproduces the queuing dynamics and switching overhead of cyclic token-passing networks and is validated against the corresponding analytical queuing models. The exhaustive and limited disciplines reach mean relative errors of 1.05% and 1.97% against theory, respectively, which we consider a good match. The sensitivity analysis over network size (M) and traffic load (ρ) shows that exhaustive polling consistently keeps average waiting times lower than limited polling. In the asymmetric traffic scenario, exhaustive service keeps fairness almost untouched (Jain's Index ≈ 0.999), while limited service causes a clear latency penalty for the high-load station.

2. Project Context and Objectives

The goal of this project is to simulate and compare polling systems under exhaustive and limited service strategies through a discrete-event model. We focus on system stability and the expected waiting time $E[W]$, both for standard symmetric configurations and for asymmetric ones. Specifically, we aim to: (i) validate the simulator against analytical queuing theory, (ii) study how $E[W]$ is affected by the traffic load ρ and the number of stations M , and (iii) measure fairness across stations when the traffic distribution is not uniform.

3. Methodology

The simulator is built around a Python discrete-event core that processes packet arrivals and departures while keeping track of each station's queue and the cyclic state of the polling server (Figure 1). We swept the traffic load over $\rho \in [0.1, 0.7]$ and the number of stations over $M \in [5, 160]$. For each configuration we computed the global average waiting time $E[W]$, the per-station waiting time $E[W_i]$, the relative error with respect to the theoretical formulas, and Jain's Fairness Index. Two sets of scenarios were run: a uniform traffic baseline, and an asymmetric extension where one station (the "hot" station) receives a higher arrival rate than the rest.

Figure 1: Block diagram of the simulation framework

4. Results and Discussion

4.1 Baseline Validation and Theoretical Alignment

The simulation output matches the theoretical models closely. The relative error stays below 2% for low values of ρ in both disciplines. The largest deviations appear only at high traffic load ($\rho = 0.7$) combined with large station counts, which is expected: as the system approaches saturation, the variance of the waiting time grows and a finite number of simulated events is not enough to fully average it out.

Table 1: Simulation Relative Error Summary

Discipline	Mean Error (%)	Median Error (%)	Max Error (%)
Exhaustive	1.05	0.93	5.27
Limited	1.97	1.69	15.00

4.2 Parameter Sensitivity (M and ρ)

Plotting $E[W]$ against the traffic load shows that both disciplines grow exponentially as $\rho \rightarrow 1$, as expected from queuing theory. However, limited service grows noticeably faster because the server is forced to switch to the next station after serving a single packet, even if the current queue still has a backlog. We also verified that $E[W]$ scales linearly with the number of stations M . At high load ($\rho = 0.7$), the gap between disciplines becomes more visible: as shown in Table 2, for $M = 10$ and $\rho = 0.7$, limited polling roughly triples the waiting time of exhaustive polling.

Table 2: Validation of Baseline (Sub-sample $M=10, \rho=0.7$)

Discipline	Theoretical $E[W]$	Simulated $E[W]$	Absolute Error	Relative Error (%)
Exhaustive	15.90	16.18	0.38	2.37
Limited	47.03	48.60	1.99	4.23

4.3 Asymmetric Traffic and Fairness Analysis

To test how each policy reacts to non-uniform traffic, we introduced a single high-demand station into the network. Table 3 shows that the two disciplines behave very differently in this scenario. Exhaustive service keeps the waiting time roughly the same across all stations (≈ 11.7) and achieves a Jain’s Fairness Index of 0.999, since each visit to a station is allowed to last as long as needed to clear its queue. Limited service, on the other hand, gives every station the same number of served packets per cycle regardless of its load, which is fair in terms of service opportunities but not in terms of delay: the hot station’s waiting time rises to 55.34, against 19.48 for the normal stations, and the fairness index drops to 0.821.

Table 3: Extension Fairness Summary

Discipline	Hot E[W]	Station	Normal E[W]	Mean	Max/Min Ratio	Jain's Index
Exhaustive	11.12		11.76		1.09	0.999
Limited	55.34		19.48		3.03	0.821

5. Conclusions and Recommendations

1. Model validity: the discrete-event simulator reproduces the analytical results with good accuracy, with a median relative error below 1.7% across all the configurations tested.
2. Performance trade-off: exhaustive service consistently gives lower average delay than limited service. This gap widens as the traffic load ρ and the number of stations M increase.
3. Robustness to asymmetric traffic: exhaustive polling handles a high-demand station well, without losing fairness, while limited polling does not.
4. Recommendation: exhaustive polling is the better choice for networks with dynamic or heterogeneous traffic, since it avoids starving high-load stations. Limited polling is only preferable in homogeneous environments where a strict, predictable bound on the token-holding time per station is required.
5. Future work: extending the analysis to k-limited or time-limited disciplines would help quantify the trade-off between the bounded service guarantees of 1-limited polling and the throughput efficiency of exhaustive polling.

